side to engage gear 7 in the main shaft as shown in Figure. Now the rotational power is transmitted to the main shaft through the dog clutch. Now the speed is low. The power transmission from the clutch shaft to the output shaft is by gear1-2 is means rotational power is transmitted to main shaft by 1-2-4-7-D_1.

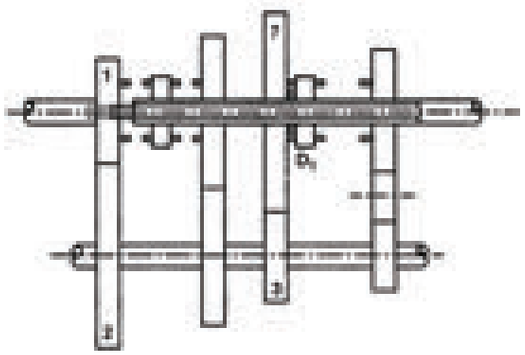


Figure 3.5.2.1 Power transmission in first gear

Second gear

For obtaining the second gear, by actuating the gear shift lever, the dog clutch D_2 is moved towards right as shown in Figure to engage the gear 8 in the output main shaft. Now the rotational power is transmitted to the main shaft through the dog clutch D_2. Now the speed of the main shaft is higher than the speed of the first gear. The power transmission from the clutch shaft to the output shaft is through the gears1-2-3-8-D_2.

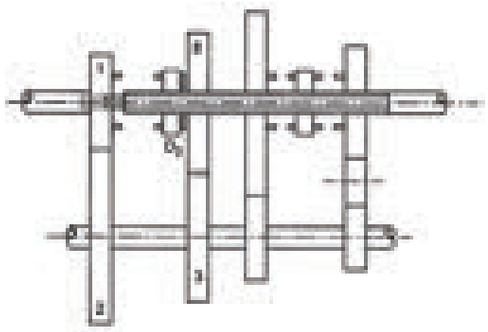


Figure 3.5.2.2 Power transmission in second gear

Third gear

To achieve the third gear the dog clutch D_2 is moved left to engage with the clutch shaft gear directly. Now the output shaft rotates at the speed same as the clutch shaft speed. The rotational power is transmitted to the main shaft directly through the gear1-D_2.

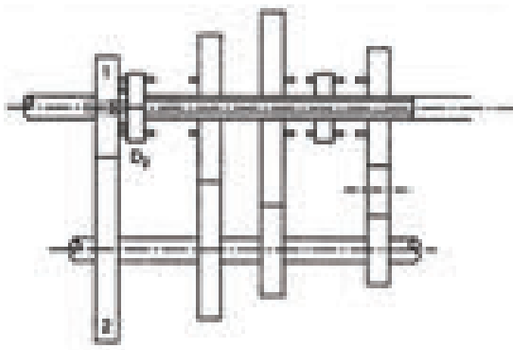


Figure 3.5.2.3 Power transmission in third gear

Reverse speed

To obtain the reverse gear, the dog clutch D_1 is actuated to move right towards the reverse gear 6 to engage as shown in Figure 3.5.2.4. As the idler gear is placed between the gears 5 in the lay shaft and 6 in the main shaft the idler gear changes the direction of rotation of the gear 6 in the main shaft. Now the vehicle moves in the reverse direction. The power transmitted from the cluth shaft to the main shaft is through gears 1-2-5-Idler-6-D_1.

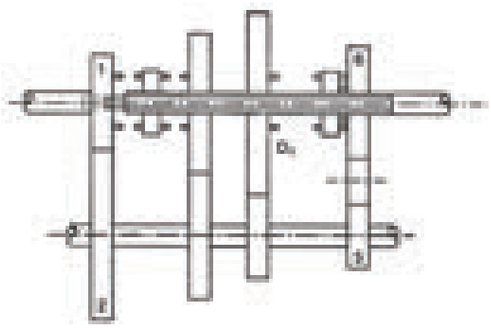


Figure 3.5.2.4 Power transmission in reverse gear

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